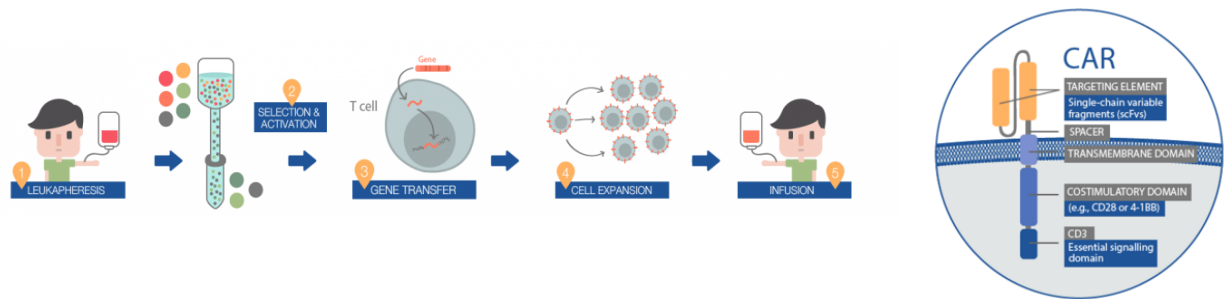


Comparing ODE Models of Chimeric Antigen Receptor T cell therapy

Biological Background:

- Chimeric antigen receptors (CAR) are engineered fusion proteins that target T cells to a specific antigen present on tumor cells to generate an antitumor immune response.
- The CAR protein contains an extracellular domain which binds to a surface marker expressed on tumor cells (and ideally not healthy cells), making them specific. It also includes an internal co-stimulatory domain which allows the engineered T cells to mount an immune response after binding to their target antigen, bypassing the requirement of additional, external costimulation inherent to endogenous T cells.
- Brief summary: A patient's T cells are isolated from a blood sample. A viral vector is used to insert the CAR gene into their T cells, which are then expanded ex vivo. Patients receive a round of lymphodepleting chemotherapy to reduce their endogenous T cells population and thereby make room for the CAR T cells to expand. Finally, CAR T cells are infused into the patient.



- The earliest CAR T cell technology was actually developed to try to cure HIV; however, this has proven a challenging area of application due to the low levels of antigen expression in controlled HIV infections (among other reasons).
- The first success came in treating blood cancers. CAR T cells targeting CD19, which is expressed on malignant B cells, were found to have potent activity in early phase clinical trials involving patients with relapsed and/or refractory B cell malignancies over a decade ago now.
- Long-term follow-up data indicate that CD19-targeted CAR T cells are likely to be curative for a subset of patients with B cell lymphomas.
- CD19-targeted CAR T cells are now approved for relapsed and/or refractory B cell lymphoma and B cell acute lymphoblastic leukaemia, and B cell maturation antigen-targeted CAR T cells are approved for relapsed and/or refractory multiple myeloma.
- CAR T cell therapies have become an important component of the treatment landscape for a range of haematological malignancies, and the indications for these therapies continue to expand to earlier lines of treatment.

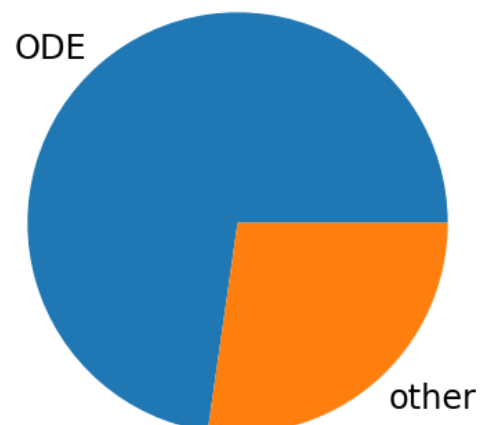
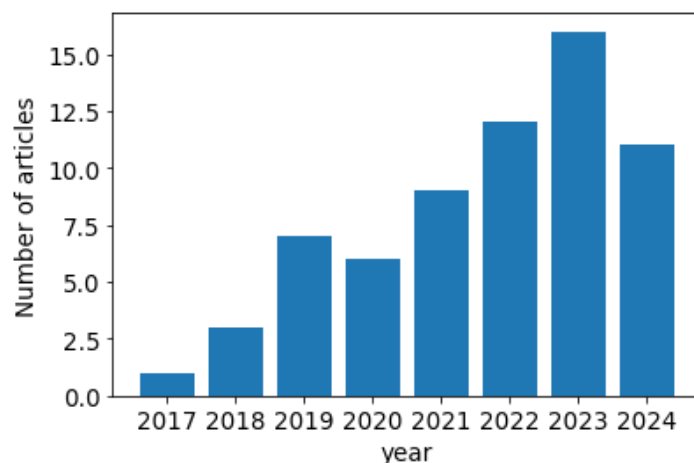
Disease	Target	Response rate	FDA-approved drugs
Large B-cell Lymphoma	CD19	40-74%	axicabtagene ciloleucel (2017) Tisagenlecleucel (2018) lisocabtagene maraleucel (2021)
B-cell Acute Lymphocytic Leukemia	CD19	71-81%	Tisagenlecleucel (2017) brexucabtagene autoleucel (2021)
Multiple myeloma	BCMA	73-98%	idecabtagene vicleucel (2021) ciltacabtagene autoleucel (2022)

HYPERLINK
<https://www.nature.com/articles/s41571-023-00754-1> *Cappell and Koschieder 2023

- There is also a lot of interest in extending this technology to treat other malignancies and other diseases, including HIV.

Mathematical Modeling:

- Mathematical modeling of CAR T cell therapy is a growing area of work. I queried google scholar for articles with “CAR T cell” and either “mathematical” OR “quantitative” OR “math” OR “model” in the title to generate the bar chart below.
- Most models are ODE-based. A subset of these have been fit to data.
- Existing studies address diverse scientific goals:
 - Optimize design or manufacturing process
 - Elucidate CAR T-cell proliferation and/or killing mechanisms
 - Predict patient outcomes
 - Suggest optimal treatment plans
- Increasingly there is data tracking CAR T cell counts and tumor size, even CAR T cell in vivo imaging tracking location of CAR T cells so I think this is an exciting area of application for data driven modeling.



Open Questions/Challenges:

- CAR T cell treatment can fail if **tumor heterogeneity** leads to antigen escape. We could build a model that includes a resistant subpopulation and look into evolutionarily informed treatment plans with pulsed CAR T cell therapy. (To manage exhaustion/toxicity people have developed CAR T cells with on/off switches).
- For solid tumors, achieving **adequate T cell trafficking to the tumor site and tumor infiltration** is challenging. We have a simple spatial model for local delivery which we could use to model additional treatment plans including multiple doses, tumor resection + local delivery, time-release delivery.
- **Managing toxicity, minimizing risk of post-treatment infections.** Consider modeling combinations of pre-conditioning + CARs + drugs for managing CRS.
- **Immunosuppressive tumor microenvironment.** Include more careful model of tumor-immune interactions with the endogenous immune system and CAR T cells including phenomena like exhaustion, bystander killing.
- What is the optimal balance of CD4 vs CD8 CAR T cells, memory vs naïve etc.

Motivation:

Before diving into these questions, it would be great to go back to the collection of ODE models that have been developed for CAR T cell therapy, and calibrate a subset of them to data on tumor size and CAR T cell count from B cell malignancies. This could serve as a literature review + model comparison. It would allow us to pick a good mechanistic model for future work by identifying which model (or collection of models) best explain available data.

Relevant CAR T cell Model Reading:

- Simple model fit to in vitro data: Sahoo et al. 2020
https://royalsocietypublishing.org/doi/full/10.1098/rsif.2019.0734?rfr_dat=cr_pub+_pubmed&url_ver=Z39.88-2003&rfr_id=ori%3Arid%3Acrossref.org
- Our model not fit to data, which I would like to validate against data: Owens and Bozic 2021 <https://link.springer.com/article/10.1007/s11538-021-00869-5>
- Liu et al. 2020 digitized data on CAR T cell kinetics from 217 patients across 7 trials and compared kinetic parameters when stratified by response status, patient populations, and tumor types. They used monolix to fit their piecewise model—we could take a similar approach with mechanistic models.
<https://ascpt.onlinelibrary.wiley.com/doi/full/10.1002/cpt.2040>.
- Liu et al. 2022 (different Liu than above!) I haven't read this in detail yet, but they provided the most ready-to use data set which has both tumor and CAR T cell data over time. "Real-time CAR T-cell and tumor burden data of 209 patients were collected from clinical studies and standardized with unified units in bone marrow."
[Liu et al. 2022 Computational model of CAR T-cell immunotherapy dissects and predicts leukemia patient responses at remission, resistance, and relapse.](#)
- More recent model accounting for different CAR T cell phenotypes fit to clinical data: [Kirouac et al. 2023](#), and [a follow up from the same author in 2024](#)

Available Data:

- CAR T cell kinetics + patient response: Liu et al. 2020 developed a cellular kinetic model to retrospectively **characterize CAR-T kinetics** in 217 patients from 7 trials and compared CAR-T kinetics across response status, patient populations, and tumor types. Model consists of linear kinetics applied in phases. They track effector and memory subsets of T cells. The data they used is available as supplemental info here: <https://ascpt.onlinelibrary.wiley.com/doi/full/10.1002/cpt.2040>.
 - o One paper they cite is the OG Tisagenlecleucel kinetics: <https://ascpt.onlinelibrary.wiley.com/doi/10.1002/psp4.12388>
 - o Last page of the supplement has Tisagenlecleucel + tumor data: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7013261/>
- CAR T cell kinetics and tumor data: [Liu et al. 2022 Computational model of CAR T-cell immunotherapy dissects and predicts leukemia patient responses at remission, resistance, and relapse](#). Data is on their github page! <https://github.com/Lunan12/CAR-T-model>
- Other models fit to data from both CAR T cells and tumor, which we could request/digitize:
 - o Kimmel et al. zuma data: https://royalsocietypublishing.org/doi/full/10.1098/rspb.2021.0229?rfr_dat=cr_pub++0pubmed&url_ver=Z39.88-2003&rfr_id=ori%3Arid%3Acrossref.org
 - o Model fit to data including endogenous T cells: <https://aacrjournals.org/cancerrescommun/article/2/11/1532/711340/Mechanistic-Modeling-of-the-Interplay-Between-Host-Effector-Memory-Exhausted-cell-model>. (Kirouac et al) We could digitize data ourselves from the same source, which is Figure 1 here: <https://www.nature.com/articles/s41591-018-0010-1>
 - o Another recent model from PKPD perspective to consider: <https://www.nature.com/articles/s41540-024-00355-3?fromPaywallRec=false>
 - o Ahmed et al. fit to in vitro data from cell assays + clinical data <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10508581/>
- Look at models fit to data in Nukula et al. review—particularly if individual-level data has CAR and tumor:
 - o Nukala et al: <https://arxiv.org/pdf/2101.05359>
 - o Tserunyan et al. is a little less helpful, but includes some more recent papers.
- In vitro experiment elucidating killing mechanism: <https://www.nature.com/articles/s41419-024-06461-8>
 - o And a relevant modeling paper: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9983577/>

Data Fitting Options:

- Individual-based approach

- Minimize RSS between data points and model prediction for each individual
- Does not incorporate information about parameter values across individuals
- Assumes gaussian distributed noise on each data point.
- Use built-in function in python, matlab, R or other
- Population-based approach
 - Population mixed effect model. Measurement from individual i at timepoint k modeled as $y = f(t, \theta) + \epsilon$, where f represents the solution of the ODE model for the relevant state variable, θ is the parameter vector for individual i , and $\epsilon \sim N(0, \sigma^2)$ is the measurement error for the data.
 - We have a statistical model that includes the mechanistic model, an error model, and any possible correlations/covariates in the population
 - Use monolix